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A comparison of imputation methods for categorical data

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ABSTRACT

Objectives: Missing data is commonplace in clinical databases, which are being increasingly used for research. Without giving any regard to missing data, results from analysis may become biased and unrepresentative. Clinical databases contain mainly categorical variables. This study aims to assess the methods used for imputation in categorical variables.

Materials and methods: We utilized data extracted from paper-based maternal health records from Kawempe National Referral Hospital, Uganda. We compared the following imputation methods for categorical data in an empirical analysis: Mode, K-Nearest Neighbors (KNN), Random Forest (RF), Sequential Hot-Deck (SHD), and Multiple Imputation by Chained Equations (MICE). The five imputation methods were first compared by accuracy of predicting the missing values. Next, the imputation methods were compared by predictive accuracy of the outcome variable in four classifiers. The consistency of performance of imputation methods across different levels of missing data (5%–50 %) was assessed by Kendall's W test.

Results: KNN imputation had the highest precision score at levels (5%–50 %) of MCAR missing data. At lower proportions of missing data (5 %, 10 %, 15 %, 20 %), RF imputation had the second-highest precision score. SHD imputation had the worst precision at all levels of missing data. In the prediction of the outcome, the methods performed differently at all proportions of missing data in the four classifiers. Even though KNN imputation was the best method in predicting the missing values, it did not consistently enhance the predictive accuracy of the classifiers at all levels of missing data. Our findings show that a high precision score of an imputation method does not translate into higher predictive accuracy in classifiers.

Conclusions: KNN imputation is the best method in predicting missing values in categorical variables. There is no universal best imputation method that yields the highest predictive accuracy at all proportions of missing data.

1. Introduction

Missing data is commonplace in quantitative research [1]. The prevailing practice to handle missing data is the use of ad-hoc methods such as using only complete records in analysis and discarding any record with even one missing value (listwise deletion) [2,3]. This, however, can lead to significant loss of data, inflated standard errors, loss of power and misleading conclusions in analysis [4–7].

Most machine learning algorithms, including classification algorithms, cannot work on incomplete datasets, and hence require some treatment of missing data beforehand, including deleting incomplete records or imputation [8]. Using only complete cases, however, has shown to result in low predictive power of classification algorithms [7,9–11]. Rather than discarding incomplete cases, researchers now have a myriad of imputation methods to select from to better handle the problem of missing data, an area that is constantly evolving [12–16].

Clinical databases are being increasingly used for developing prediction models for morbidities. This is due to availability of large volumes of relevant data collected by healthcare professionals. Such data would be expensive to collect in surveys and even subject to errors due to recall bias. However, these data usually face the problem of missingness since it is not ideally recorded for the intent of research; data that is not needed during a patient's visit is not recorded [6,17]. Analysis on clinical datasets using only complete records could lead to false conclusions [6,18].

Most, if not all, variables in epidemiological studies are categorical in nature to ease medical decision process [19]. Yet, most studies done on imputation methods have focused on either numeric variables or only one type of categorical variables, such as only binary variables, or only multinomial variables [9,11,20,21]. In reality, however, clinical

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databases usually consist of all types of categorical variables; binary, ordinal and multinomial. The objective of this study is to determine the optimal imputation method for categorical data at varying levels of missingness.

2. Theoretical background and related work

With regard to a threshold of missing data beyond which quality statistical inference is not possible, there is lack of agreement in literature. Bennett has stated that analysis will be biased if more than 10 % of data is missing [22]], whereas Schafer asserted that inference would be misleading if more than 5 % of data was missing [23]]. Tabachnick, Fidell and Ullman [24]], on the other hand, argue that the missing data mechanism, rather than the extent of missing data, has a greater influence on statistical analysis.

2.1. Missing data mechanisms

Little and Rubin [25] have identified three missing data mechanisms; Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR). These mechanisms have been discussed below.

Missing Completely at Random (MCAR): Data is said to be MCAR when the probability of missing depends neither on observed, nor on missing data, and is the same for all cases. Use of listwise deletion in this case does not lead to biased estimates but may significantly reduce the sample sizes for analysis, hence resulting in increa3.sed standard errors [1,6,7,18].

Missing at Random (MAR): Here, the probability of missingness in one variable depends on observed values in another variable. It, however, does not depend on unobserved values. Rubin has stated that the effects on parameter estimates are minimal under violation of the MAR assumption [26].

Missing Not at Random (MNAR): This is the most serious case of missingness where the probability of missingness depends on the missing value itself. This happens usually because the question could have been too sensitive for the respondent to answer. Statistical analysis in the presence of MNAR data can have severe effects on parameter estimates [16,26]. Rubin, Stern, and Vehovar [27] emphasized the necessity of adding covariates in the imputation process to make the MAR assumption more plausible.

Many published studies use ad-hoc methods for handling missing data [1–3]. However, these methods require assumptions about the data which rarely hold in practice [12,13]. It is, therefore, important to first identify the mechanism of missing data before selecting a method to treat missingness. Failure to do so could have serious impact on parameter estimates including biased and/or inefficient estimates [26, 28].

2.2. Methods of handling missing data

Researchers treat missing data in either one of two ways; they either ignore the missing values or they impute the missing values using an imputation technique.

Ignoring Missing Values: In this approach, cases with incomplete data are omitted. There are two methods under this approach; listwise deletion and pairwise deletion. Listwise deletion uses only complete cases in analysis and ignores all cases with missing values in even one variable. It is the default method used to handle missing data in most statistical software. However, it yields unbiased estimates only if data is MCAR [16] but high standard errors due to significant loss of data (50 % on average) [7,29]. Listwise deletion gives the lowest predictive accuracy in machine learning algorithms [7,9,10]. In pairwise deletion, analysis is performed on all cases available in the variables of interest. This method retains more data than complete case analysis but ends up having different sample sizes for different variables [30].

Imputing Missing Values: In this approach, a missing value is replaced by a plausible value(s) using a statistical or machine learning technique [9,26]. Imputation ensures that the sample size is maintained, hence giving more accurate estimations of population parameters and better classification and predictive accuracy in data mining and data analysis techniques [7,9,11]. The two approaches under imputation are single imputation and multiple imputation. In the case of single imputation, only one plausible value is imputed for a missing value and then analysis is performed on the complete dataset. The drawback of single imputation is that analysis will treat the imputed value as the true value. This will underestimate the uncertainty of the estimates and hence overstates precision [16]. Multiple imputation overcomes this drawback by incorporating the uncertainty of the validity of the imputed values by providing multiple plausible values for each missing value [13].

2.3. Imputation methods for categorical data

Not all imputation methods can work for categorical data, for instance mean imputation, median imputation, predictive mean matching, etc., work for only numeric variables. Additionally, some imputation methods for categorical data work only on binary data, for instance multiple imputation based on Logistic regression. In empirical data, however, categorical variables are usually of all three types; binary, multinomial, and ordinal. Imputation methods that can handle all types of categorical variables have been discussed below.

Mode Imputation: This is one of the simplest and fastest method of dealing with missing data, where missing values are replaced with the mode of a categorical variable. It, however, poses the risk of over-representation of one category in a variable leading to underestimation of variance in the variable. Low variation in X may lead to over optimistic results. Mode imputation requires data to be MCAR; it results in biased estimates otherwise [16].

K-Nearest Neighbors (KNN) Imputation: The K neighbors (cases) that are most similar to the case with missing value are selected and the corresponding value is imputed in the missing value. Similarity is measured using a distance, for instance Hamming distance for categorical variables. KNN imputation is not affected by missing mechanism; it can be used when data is MCAR, MAR or MNAR. However, it requires prior knowledge of tuning parameter K [31].

Sequential Hot-Deck (SHD) Imputation: The data is first sorted on one or more background characteristic. The algorithm then makes only one pass over the records in the dataset in order and imputes each missing value in a record, by the latest previously encountered observed value from another record that is similar to the missing record with respect to one or more characteristics shared by both records. It is one of the fastest and most commonly used imputation in research. It, however, requires the data to be MCAR [32].

Random Forest (RF) Imputation: A Random Forest is first trained on observed values. Next, missing values in a variable are predicted using other variables and the trained random forest. This process is repeated iteratively until a stop criterion is fulfilled. Unlike KNN imputation, there is no tuning parameter, and hence there is no need for any trial and error to find the best parameter [14].

Multiple Imputation using Chained Equations (MICE): MICE imputation uses regression to predict m plausible values per missing value and, hence, generates m datasets. After imputation, any choice of analysis, such as regression, is performed on each imputed dataset. Next, the estimates and standard errors generated in analysis from each imputed dataset are combined using Rubin's rules [33]. MICE is a popular choice for imputation due its capability of imputing all variable types. This is because it can use the appropriate univariate imputation model for each variable type, including continuous and categorical (both ordered and unordered). For categorical variables, the following univariate imputation models are used: logistic regression for binary variables, polytomous logistic regression for multinomial variables, and proportional odds model for ordinal data. Predictive mean matching, classification

and regression trees, and random forest univariate imputation models are for all types of categorical variables. However, MICE loses efficiency with increased variables, especially categorical variables due to creation of dummies which are used as predictors for variables with missing values [16].

2.4. Use of clinical databases in research

Administrative health records exist as electronic records (EHR) or paper records. Many developed countries have adopted EHR systems whereby information is recorded and stored using electronic devices. This reduces the occurrence of missing data due to fields that cannot be bypassed by data entrants during data collection. Paper medical records, on the other hand, do not have such a provision and the healthcare attendee can skip over any field, leading to increased incomplete data. Studies have shown that paper medical records are harder to retrieve when needed and have lower rates of completeness as compared to electronic medical records [34,35].

Many studies done on maternal outcomes in developing countries have utilized facility-based health records [36–46]. Most of these studies do not mention missing data or they use only complete cases. It is important to note that most, if not all, of the variables used in these studies are categorical in nature. Most demographic factors used in these studies are already categorical, and factors that are continuous are usually categorized. This is in line with a study done on categorization of continuous risk factors in epidemiological publications. The study found that continuous variables were categorized in 86 % of the articles surveyed [47]. Categorization is usually done to aid in medical decision making process where results are perceived as either normal or abnormal and hence necessary interventions can be applied [19]. The question, then, would be: what method of imputation would perform best on categorical data?

2.5. Related work

Comparison of imputation methods is not a new concept in literature. A few studies have compared methods for handling missing data in Electronic Health Records (EHR) from developed countries. For instance, Jerez et al. [10] compared machine learning and statistical imputation methods for imputing missing data in an EHR in Spain. For the prediction of breast cancer, imputation done using machine learning methods performed better than imputation done using statistical methods. Bounthavong, Watanabe, and Sullivan [48] compared complete case analysis and multiple imputation for handling missing data in an EHR from the United States. The use of multiple imputation yielded the same conclusions in regression analysis as those based on complete cases.

Other studies have used a more experimental approach to compare imputation methods, whereby missing data is first artificially implanted in complete datasets. Kyureghian, Capps, and Nayga [49] examined both single and multiple imputation methods. In simulated datasets, they induced missingness and compared imputation methods by measuring accuracy of predicting the missing values and the bias in parameter estimates from regression. Overall, multiple imputation methods yielded less biased parameter estimates and gave better predictions of the missing values. Twala [50] artificially simulated three proportions (15 %, 30 %, and 50 %), patterns, and mechanisms of missing data in 21 complete datasets obtained from the UCI repository. He compared seven missing data techniques and found that listwise deletion was the worst technique while multiple imputation was the most superior approach in handling missing data, especially if a dataset contains large proportions of missing data. Jadhav, Pramod, and Ramanathan [20] utilized seven numeric datasets from the UCI repository to compare seven data imputation methods. At all levels of missing data (10 %, 20 %, 30 %, 40 %, and 50 %), they found that KNN imputation had the lowest Normalized Root Mean Square Error (NRMSE). Penone et al. [21] used a nearly complete trait dataset to compare KNN, MICE, and RF imputation. They artificially removed 10 %–80 % of data and using NRMSE as a measure of imputation performance, they found that KNN imputation had the largest errors.

As earlier discussed, datasets used in clinical research have mostly categorical variables. A few studies have looked into imputation methods for categorical variables. For example, using two machine learning benchmark datasets, Poulos and Valle [51] examined imputation methods for missing categorical data. They compared classifiers trained on non-imputed and imputed data at varying levels of additional missing-data perturbation. They showed increased predictive accuracy of classifiers trained on imputed data with perturbation. Ghorbani and Desmarais [9] compared imputation methods for binary data. They induced between 5 % and 50 % ranges of missing data in fourteen datasets from various repositories and assessed the performance of the imputation methods based on classification accuracy of four different classifiers. Their results showed that MIEM outperformed the other methods. Ramosaj and Pauly [52] compared a few methods of enhancing RF imputation for continuous, categorical as well as mixed-type data according to increasing imputation accuracy.

The surveyed literature shows that there is no one single imputation method that yields the best results across all datasets and datatypes. There is an abundance of studies done to compare imputation methods but most of these have been done in the context of EHR, developed countries and numeric datasets. This study aims to address the gap in literature by examining imputation methods for categorical variables in paper-based medical records from a developing country.

3. Methodology

This section describes the source of data, data collection, variables used in this study and steps taken in data analysis.

3.1. Data collection and description

We used data extracted from paper-based maternal health records from Kawempe National Referral Hospital which mainly provides antenatal, intrapartum, and postnatal care. Data on Severe Maternal Morbidities (SMMs) and their predictors were captured using a tool designed in Open Data Kit (ODK). Only records from January 2017 to January 2021 were considered. Maternal records without ANC cards were not used as the ANC cards contained most of the socio-demographic and medical information required in this study.

Data was captured with the support of six research assistants with medical backgrounds. Quality control included training the research assistants prior to data collection, pretesting the tool, and supervision. Starting from most recent and going backwards, each record was checked for a category of interest (eclampsia, hemorrhage, obstructed labor, sepsis). These formed the cases; i.e., women with an SMM. We used a case-control ratio of 1:1; for every file with an SMM, a file without an SMM in the same day was selected as a control. A total of 5036 records were captured; 2518 cases and 2518 controls. Table 1 summarizes the variables considered in this study.

The response variable in this study was incidence of SMM. This was deduced from indication of either eclampsia, hemorrhage, sepsis or obstructed labor in the delivery notes. A woman with any one of these SMMs was recorded as having an SMM and a woman who did not develop any one of these complications during pregnancy or childbirth was recorded as not having an SMM.

3.2. Data analysis

Analysis was done in RStudio in the following steps: first, incomplete observations were deleted in the original dataset such that only complete observations remained (hereafter referred to as the *complete* dataset). The performances of the imputation methods were then

Table 1
Variables adopted in the study.

Variable	Categories	Variable Type
Religion	Catholic, Muslim, Others, Protestant	Multinomial
Pre gestation diabetes	Yes, No	Binary
HIV serostatus	Yes, No	Binary
Pre gestational hypertension	Yes, No	Binary
Asthma	Yes, No	Binary
Previous blood transfusion	Yes, No	Binary
Multiple gestation	Yes, No	Binary
Previous D&C	Yes, No	Binary
Previous ectopic pregnancy	Yes, No	Binary
STD	Yes, No	Binary
Previous D&E	Yes, No	Binary
Familial diabetes	Yes, No	Binary
Familial hypertension	Yes, No	Binary
Referral	Yes, No	Binary
Previous cesarean	Yes, No	Binary
Pre delivery weight	≤55, >55	Binary
District	Kampala, Other	Binary
Age group	<20, <35, ≥35	Ordinal
Employment status	Employed, Self-employed, Unemployed	Multinomial
Marital status	Married, Unmarried	Binary
Education level	Primary, Secondary, Post-secondary	Ordinal
Family planning method	Implant, Injection, None, Oral, Other	Multinomial
Gravidity	Primi, Multi, Grand	Ordinal
Parity	Nulliparous, <4, ≥4	Ordinal
Number of ANC visits	One, Two, Three, Four or more	Ordinal
Trimester at first ANC visit	Nulliparous, <4, ≥4	Ordinal
Previous abortion	Yes, No	Binary
Previous stillbirth	Yes, No	Binary
Previous preterm birth	Yes, No	Binary
Previous ENND	Yes, No	Binary

compared in two ways; precision score and accuracy of prediction of outcome variable in four classifiers after imputation.

3.2.1. Comparison of imputation methods by precision score

In the first approach, missing values were injected using the MCAR mechanism into copies of the complete dataset (except in the dependent variable) in the following seven proportions: 5 %, 10 %, 15 %, 20 %, 30 %, 40 %, and 50 % as guided by previous studies on imputation methods [22,32]. The MCAR mechanism is explored in this study due to its assumption that the distribution of the missing data and observed data are the same, and valid inference can be drawn from the available data [11,53]. This was done using the *missMethods* library in RStudio. This step yielded seven datasets (hereafter referred to as *missing* datasets) with varying proportions of missing values. Subsequently, five methods; Mode imputation, MICE imputation, RF Imputation, SHD imputation and KNN imputation, were used to impute the missing values in each of the seven missing datasets to form complete datasets (hereafter referred to as *imputed* datasets). This resulted in 7*5 = 35 datasets. These methods were selected based on their ability to impute missing values in all three types of categorical variables simultaneously. The following RStudio libraries were used for their respective imputation methods: Mode imputation was done using *missMethods* library, MICE imputation was done using *MICE* library, KNN imputation and SHD imputation were done using *VIM* library, and RF imputation was done using *missForest* library. In MICE imputation, the Random Forest univariate imputation method was used, based on Van Buuren's recommendation for categorical variables with events of low frequencies, which is common in medical data [16]. Based on Rubin's study [13] which concludes that there is little advantage in generating more than a few datasets (3–10), we used m = 5 imputed datasets in the MICE method. In KNN imputation, we experimented with K = 5,10,15,20,25,30. K = 20 gave the highest precision and hence has been used in this study.

The accuracy of imputation was measured using precision score in all 35 imputed datasets. Precision score is the proportion of imputed values that are equal to the true values. The higher the precision score, the better the imputation method at predicting missing values. Precision score was computed using the *missMethods* library in RStudio. Because multiple imputed datasets are generated in MICE imputation, precision was computed based on the average of precision scores from all five imputed datasets. The imputation methods were ranked by decreasing precision.

3.2.2. Comparison of imputation methods by performance in four classifiers

In the second approach to compare imputation methods, the complete dataset was first split into a training (75 %) and testing dataset (25 %). Different levels of MCAR missing data (5 %, 10 %, 15 %, 20 %, 30 %, 40 %, and 50 %) were injected into *only* copies of the training set. The five imputation methods were then used to impute missing data in the seven training sets. Thereafter, four classifiers were trained in a 10-fold cross-validation process on each of the 35 training sets. The prediction of SMM was evaluated by area under the receiver operating characteristic curve (AUC) on the testing dataset. The four classifiers (R libraries) utilized were: Logistic regression, Naïve Bayes (*naivebayes*), Support Vector Machine (*e1071*), and Random Forest (*randomForest*). The imputation methods were ranked by decreasing AUC. It is important to note that the same testing set was used in all classifiers at all proportions of missing data.

3.2.3. Test statistic used to compare the performance of imputation methods

To assess the consistency of the performance of the imputation methods across all proportions of missing data, the Kendall's W test was used. The Kendall's coefficient of concordance, W, ranges from 0 to 1. The following hypotheses were formulated:

H0. there is no consistency of the rankings of the imputation methods at different proportions of missing data (W = 0)

Ha. there is consistency of the rankings of the imputation methods at different proportions of missing data (W = 1)

Fig. 1 displays the workflow adopted in this study.

4. Results

This section presents the performance of imputation methods used to impute missing data in categorical dataset at varying proportions of missingness.

4.1. Comparison of imputation methods by precision

After injecting MCAR missingness in copies of the complete dataset at 5 %, 10 %, 15 %, 20 %, 30 %, 40 %, 50 %, five methods of imputation were used in each missing dataset. This resulted in 35 imputed datasets. Precision score was used to determine the accuracy of imputation methods. Table 2 presents the precision scores of the imputation methods at 7 levels of missingness. Table 3 presents the ranking of the imputation methods by precision score.

Results in Tables 2 and 3 show that KNN imputation consistently had the highest precision at all levels of missingness. RF imputation had the second highest precision at 5 %, 10 %, 15 %, and 20 % levels of missing data. Mode imputation gave almost the same precision at all levels of missingness, and hence at highest levels of missingness (40 %, 50 %) performed almost the same as KNN imputation. SHD imputation had the lowest precision at all levels of missing data. There was consistency of rankings of different imputation methods at the different proportions of missing data (W = 0.853, p = 0.0000842). Apart from Mode and SHD imputation, all other methods' precision score reduced with increased missing data. (Fig. 2).

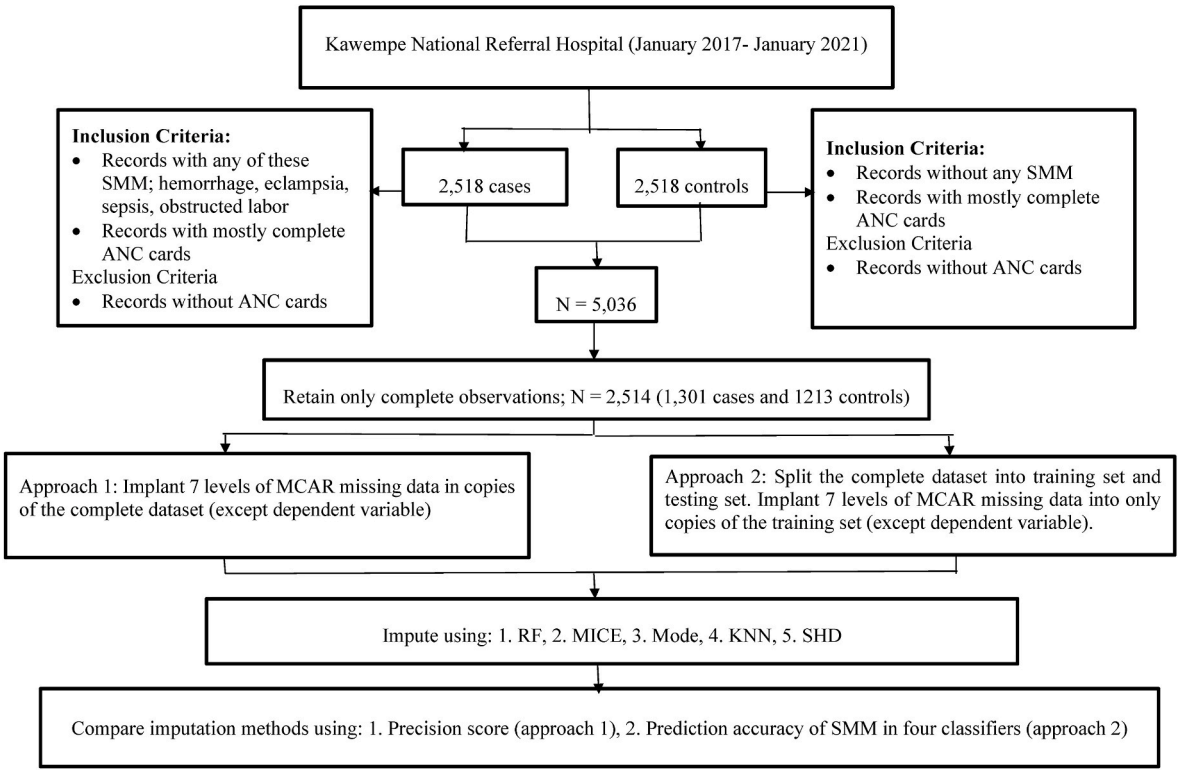

Fig. 1. Study workflow.

Table 2
Precision of imputation methods.

Imputation Method	Proportion of missing data (%)						
	5	10	15	20	30	40	50
Mode	0.8087	0.7991	0.7999	0.7909	0.7971	0.7991	0.7997
MICE	0.8042	0.7915	0.7913	0.7865	0.7824	0.7779	0.7717
SHD	0.7495	0.7398	0.7466	0.7395	0.7420	0.7395	0.7444
KNN	0.8344	0.8205	0.8178	0.8097	0.8135	0.8081	0.8041
RF	0.8272	0.8157	0.8094	0.8020	0.7802	0.7768	0.7583

Note: Precision is proportion of imputed values that are equal to the complete values. The higher the precision score, the better the imputation method.

Table 3
Rank of imputation methods by precision.

Imputation method	Proportion of missing data (%)							Rank of method by mean
	5	10	15	20	30	40	50	
Mode	3	3	3	3	2	2	2	2.6
MICE	4	4	4	4	3	3	3	3.6
SHD	5	5	5	5	5	5	5	5.0
KNN	1	1	1	1	1	1	1	1.0
RF	2	2	2	2	4	4	4	2.9
Kendall's Statistics	Chi-sq = 23.9, p = 0.0000842, Kendall's W = 0.853 (large effect size)							

Note: Ranking is ordered from highest precision to lowest, where a rank of 1 indicates highest precision and a rank of 5 indicates lowest precision.

4.2. Comparison of imputation methods by prediction accuracy in four classifiers

The complete dataset was split into a training dataset (75 %) and testing dataset (25 %). Seven proportions of missing data were injected into copies of only the training dataset using the MCAR mechanism, and subsequently imputed using the five imputation methods. Four classifiers were trained on the imputed datasets to predict incidence of SMM. The prediction accuracies of the classifiers in each imputed dataset were evaluated using AUC in the same testing dataset. Results in Tables 4–7 show the rankings of imputation methods by classification accuracy in four different classifiers at seven levels of missing data.

In Logistic regression, there was no consistency of performance in imputation methods. All five methods performed differently at different levels of missing data (W = 0.049, p = 0.849) (Table 4). In Random Forest, RF imputation overall performed better than the other methods (mean rank = 2), and KNN imputation performed the worst (mean rank = 3.7). However, Kendall's W was small (W = 0.155, p = 0.362) implying overall inconsistency of imputation methods at all levels of missing data (Table 5). In Support Vector Machine, there was no consistency in performance of imputation methods at different levels of missing data (W = 0.0531, p = 0.829) (Table 6). Contrary to its superior performance in Random Forest, RF imputation had the second worst overall performance in Naïve Bayes (mean rank = 3.6), and mode imputation performed the best (mean rank = 2.3). KNN imputation had the worst overall performance again (mean rank = 3.7) However, there was no overall consistency of the performance of imputation methods at all levels of missing data (W = 0.151, p = 0.376) (Table 7). There is major variation in the performances of the imputation methods in the different classifiers and at different proportions of missing data (Fig. 3). The performance of imputation methods becomes much more varied at

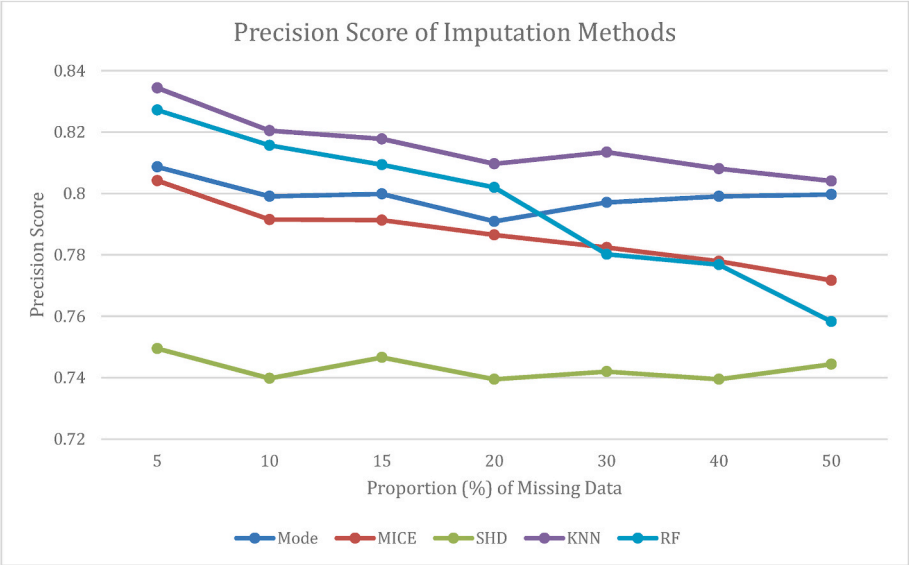

Fig. 2. Graphical comparison of imputation methods by precision.

Table 4
Rank of imputation methods by prediction accuracy in logistic regression.

Imputation method	Proportion of missing data (%)							Rank of method by mean
	5	10	15	20	30	40	50	
Mode	4	3	5	2	5	1	2	3.1
MICE	1	2	4	3	4	4	1	2.7
SHD	5	5	2	5	1	2	4	3.4
KNN	2	1	3	1	3	5	3	2.6
RF	3	4	1	4	2	3	5	3.1
Kendall's Statistics	Chi-sq = 1.37, p = 0.849, Kendall's W = 0.049 (small effect size)							

Note: ranking is ordered from largest AUC to smallest AUC, where a rank of 1 indicates largest AUC, and a rank of 5 indicates smallest AUC.

Table 5
Rank of imputation methods by prediction accuracy in random forest.

Imputation method	Proportion of missing data (%)							Rank of method by mean
	5	10	15	20	30	40	50	
Mode	2	4	3	1	4	3	4	3.0
MICE	3	2	4	5	3	4	1	3.1
SHD	5	3	5	2	1	1	5	3.1
KNN	4	5	1	3	5	5	3	3.7
RF	1	1	2	4	2	2	2	2.0
Kendall's Statistics	Chi-sq = 4.34, p = 0.362, Kendall's W = 0.155 (small effect size)							

Note: ranking is ordered from largest AUC to smallest AUC, where a rank of 1 indicates largest AUC, and a rank of 5 indicates smallest AUC.

Table 6
Rank of imputation methods by prediction accuracy in support vector machine.

Imputation method	Proportion of missing data (%)							Rank of method by mean
	5	10	15	20	30	40	50	
Mode	5	1	5	1	2	4	1	2.7
MICE	3	4	3	2	4	2	2	2.9
SHD	4	5	1	5	1	1	5	3.1
KNN	2	2	4	3	5	5	4	3.6
RF	1	3	2	4	3	3	3	2.7
Kendall's Statistics	Chi-sq = 1.49, p = 0.829, Kendall's W = 0.0531 (small effect size)							

Note: ranking is ordered from largest AUC to smallest AUC, where a rank of 1 indicates largest AUC, and a rank of 5 indicates smallest AUC.

Table 7
Rank of imputation methods by prediction accuracy in Naïve Bayes.

Imputation method	Proportion of missing data (%)							Rank of method by mean
	5	10	15	20	30	40	50	
Mode	3	1	3	1	4	1	3	2.3
MICE	2	2	2	3	5	3	2	2.7
SHD	4	3	1	2	2	2	5	2.7
KNN	5	4	4	4	1	4	4	3.7
RF	1	5	5	5	3	5	1	3.6
Kendall's Statistics	Chi-sq = 4.23, p = 0.376, Kendall's W = 0.151 (small effect size)							

Note: ranking is ordered from largest AUC to smallest AUC, where a rank of 1 indicates largest AUC, and a rank of 5 indicates smallest AUC.

larger proportions of missing data (Fig. 4).

5. Discussion

We utilized facility-based paper maternal health records to assess five imputation methods for addressing missingness in categorical data at varying proportions of missingness. The dataset utilized in this study contained binary, multinomial and ordinal categorical variables. Two approaches were used to compare the performance of: (i) Random Forest (RF) imputation, (ii) K-Nearest Neighbors (KNN) imputation, (iii) Mode imputation, (iv) Multiple Imputation by Chained Equations (MICE), and (iv) Sequential Hot-Deck imputation (SHD). The choice of these methods was based on their ability to impute missing data in binary, ordinal and multinomial categorical variables simultaneously.

To compare the five imputation methods, we first deleted all incomplete records from the original dataset. We then artificially injected seven proportions (5 %, 10 %, 15 %, 20 %, 30 %, 40 %, 50 %) of missing data in copies of the complete dataset using the MCAR mechanism. The missing data was then imputed using the five methods, yielding 35 imputed datasets. The first approach to compare the imputation methods was use of precision score, which is the proportion of imputed values that are equal to the true values. The higher the precision score of an imputation method, the better it is at estimating missing values. KNN imputation had the highest precision score at all levels of missing data. This is in line with findings from a study done on numeric variables which found that KNN imputation has the lowest Normalized

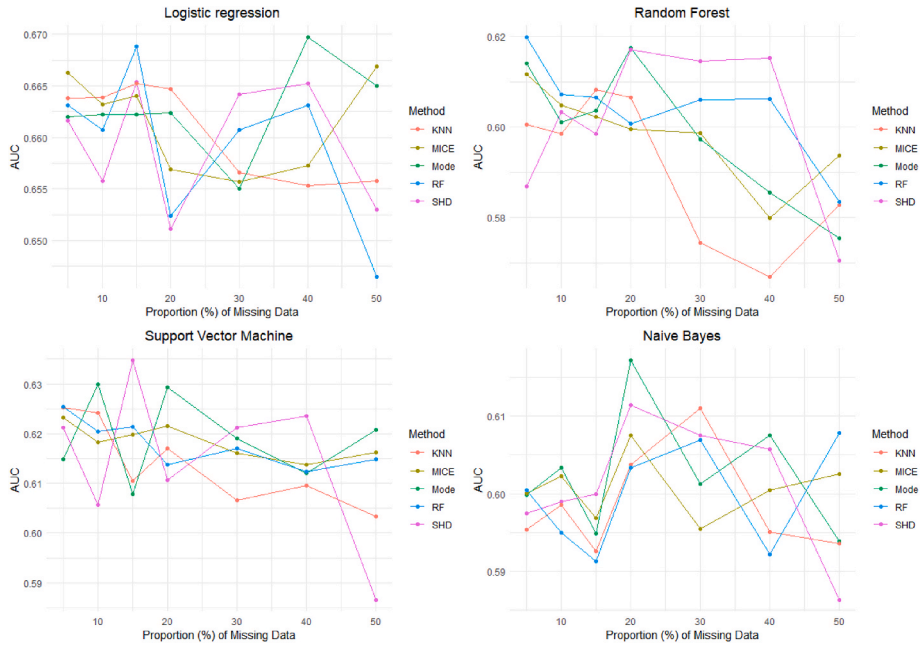

Fig. 3. Graphical comparison of imputation methods by prediction accuracy in four classifiers.

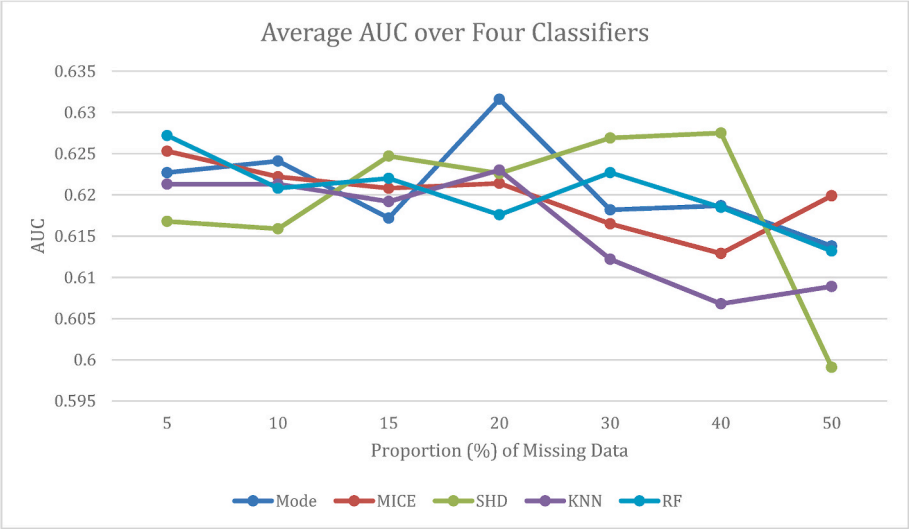

Fig. 4. Graphical comparison of imputation methods by average prediction accuracy over four classifiers.

Root Mean Square Error at all levels of missing data compared to other imputation methods for numeric data [20]. At lower proportions of missing data (5 %, 10 %, 15 %, 20 %), RF imputation had the second-highest precision score. SHD imputation had the worst precision at all levels of missing data. It is interesting to note that the precision scores for all imputation methods decreased with increasing missing data, except for Mode and SHD imputation which remained almost the same at all levels of missing data. This meant that at higher levels of missing data (30 %, 40 %, 50 %), mode imputation performed better than RF imputation whose precision score had drastically reduced. The findings suggest that KNN imputation outperforms other imputation methods in predicting missing values, irrespective of data type.

In the second approach to compare imputation methods for categorical data, the complete dataset was split into the training and testing datasets. Seven levels (5 %, 10 %, 15 %, 20 %, 30 %, 40 %, 50 %) of MCAR missing data were injected into copies of only the training dataset. The missing data was then imputed by the five imputation methods, yielding 35 training datasets. Four classifiers, namely: Logistic Regression, Random Forest, Support Vector Machine, and Naïve Bayes were trained on the 35 training sets. Prediction accuracy of the outcome variable, Severe Maternal Morbidity (SMM), was determined by area under the receiver operating characteristic curve (AUC) on the testing dataset. In the findings, there was no single imputation method that gave the highest predictive accuracy in different classifiers at all levels of missing data. This is similar to findings in other studies done on imputation methods for binary data [9] and discrete data [11]. Our findings show that a high precision score of an imputation method does not necessarily translate into higher predictive accuracy in classifiers. For instance, KNN resulted into an overall lowest predictive accuracy in Random Forest, Support Vector machine, and Naïve Bayes classifiers, yet it had the highest precision score at all levels of missing data.

6. Conclusions

KNN imputation had the highest accuracy in predicting missing values in categorical data. On the other hand, there is no universally best imputation method that yields the highest predictive accuracy across all classifiers at different rates of missing values. The findings further shows that an imputation method having a high precision score does not always mean that it will yield the highest predictive accuracy in a classifier. Even though KNN has the highest precision score across all levels of missing data, it did not end up enhancing the predictive accuracy of classifiers at all levels of missing data.

Future research focus

There is need to investigate performance of imputation methods for categorical variables in the presence of MAR and MNAR missingness.

Ethical approval

The study was conducted in accordance with the Uganda National Council for Science and Technology (UNCST) national guidelines for research involving humans as research subjects. Administrative clearance to extract data from the maternal records was obtained from Kawempe National Referral Hospital. Ethical approval for conducting the research was obtained from the Mulago Hospital Research and Ethics Committee. Finally, permission to conduct research in Uganda in accordance with the national guidelines for research was obtained from UNCST (registration number with the UNCST is HS977ES). Additionally, the study was done according to the Helsinki Declaration (1975; 2008) guidelines for medical research: (i) Names of mothers were not used; (ii) At data analysis, no individual records were reported – only aggregate findings were reported; (iii) Data was extracted from the records after discharge or death and hence study inclusion will not have an effect on the treatment; (iv) Because of the approach to be used, individual informed consent was required; (v) The data extracted has been stored safely and was used for the sole purpose of this study.

Data availability

The data utilized in this study are not publicly available and are the property of Kawempe National Referral Hospital but can be availed from the corresponding author on reasonable request.

Sources of funding

Partial funding to support data collection was received from the African Centre of Excellence in Data Science, University of Rwanda.

Consent

This was a retrospective study and only patients' records were utilized in research.

Author contribution

All authors contributed to the study's conception and design. Data collection and curation was done by SM. Formal analysis were performed by SM, RW and IK. The first draft of the manuscript was written by SM and all authors commented on, reviewed, and edited previous versions of the manuscript. All authors read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.imu.2023.101382.

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